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In [4]: import numpy as np
import matplotlib.pyplot as plt

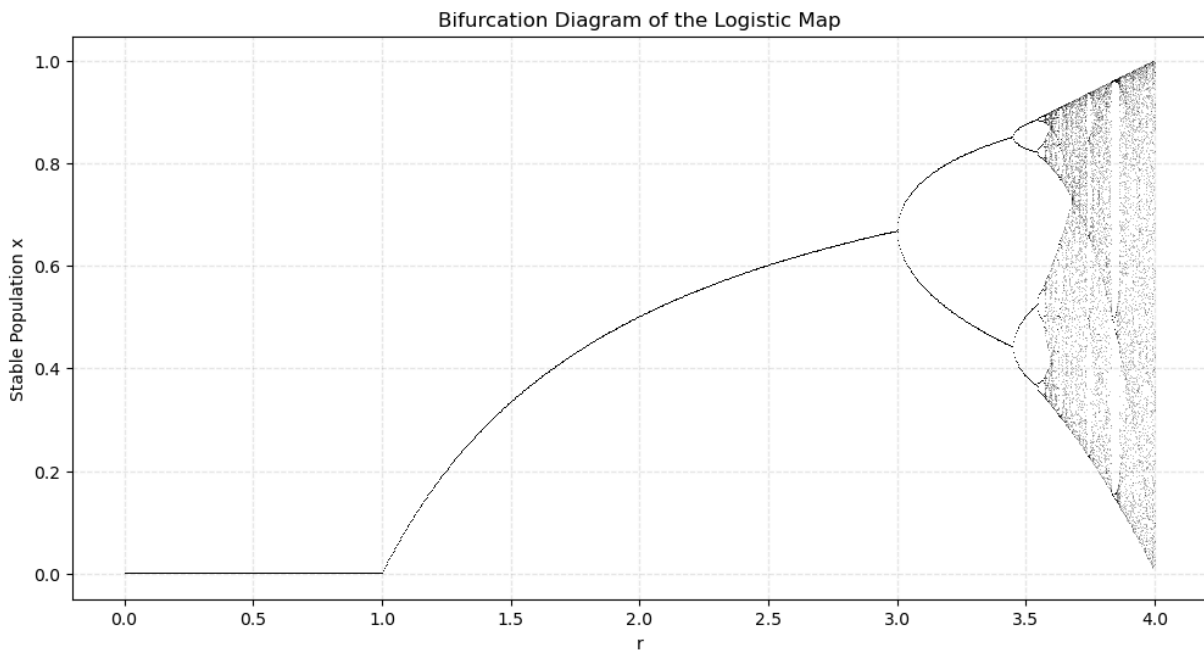
r_values = np.linspace(0, 4.0, 1000)
iterations = 1500
last = 100
x = 1e-5 * np.ones(1000)

# To store final values for plotting
r_plot = []
x_plot = []

for i in range(iterations):
    x = r_values * x * (1 - x)
    if i >= (iterations - last):
        r_plot.extend(r_values)
        x_plot.extend(x)

# Plotting
plt.figure(figsize=(12, 6))
plt.plot(r_plot, x_plot, ',k', alpha=0.25)
plt.title("Bifurcation Diagram of the Logistic Map")
plt.xlabel("r")
plt.ylabel("Stable Population x")
plt.grid(True, linestyle='--', alpha=0.3)
plt.show()

```



In []: